

BRESCIA GHEDI, ItalyWMO: **160880**Lat: **45.432N**Lon: **10.268E**Elev: **102**StdP: **100.11**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.9	-4.3	-13.8	1.1	6.1	-10.1	1.6	4.2	8.3	7.3	6.7	6.9	1.1	40	0.412

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	33.4	22.1	32.1	21.6	30.9	21.1	24.9	29.0	23.9	28.1	23.0	27.7	2.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.0	26.1	22.8	17.7	25.7	21.7	16.6	24.9	76.1	29.2	72.1	28.3	68.6	27.5	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.0	5.0	-8.3	35.3	2.1	1.5	-9.8	36.3	-11.0	37.2	-12.1	38.0	-13.6	39.0		
(5)			-8.7	26.4	2.0	2.0	-10.1	27.9	-11.3	29.0	-12.4	30.1	-13.8	31.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	2.4	4.5	9.1	13.3	18.4	22.5	24.6	24.0	19.5	14.3	8.1	3.1
	DBStd	8.39	2.72	2.90	3.21	3.04	3.06	3.14	2.39	2.74	2.88	3.19	3.41	3.10
	HDD10.0	743	237	154	53	7	0	0	0	0	0	4	74	214
	HDD18.3	2291	495	387	285	152	38	4	0	1	20	130	306	472
	CDD10.0	2097	0	1	27	106	259	376	453	434	287	137	18	1
	CDD18.3	603	0	0	0	2	39	131	194	177	56	4	0	0
	CDH23.3	5641	0	0	2	32	364	1233	1929	1661	402	18	0	0
	CDH26.7	2072	0	0	0	2	84	444	781	671	88	1	0	0
Wind	WSAvg	1.9	1.5	1.8	2.2	2.3	2.2	2.1	1.9	1.8	1.8	1.6	1.8	1.4
Precipitation	PrecAvg	896	66	59	65	65	84	82	74	87	68	92	93	56
	PrecMax	1264	155	127	159	116	151	181	177	183	209	297	155	95
	PrecMin	617	5	0	1	27	10	9	14	11	5	0	28	20
	PrecStd	181	45	35	45	27	48	47	49	52	61	80	41	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	16.2	21.6	26.7	31.2	34.2	34.7	35.7	31.2	25.3	17.4	12.2
		MCWB	8.0	9.6	11.5	15.0	18.6	22.8	23.4	21.9	20.0	17.6	14.7	9.3
	2%	DB	9.8	13.5	19.2	23.9	29.0	32.7	33.1	33.7	29.1	23.1	15.5	10.2
		MCWB	6.5	7.7	11.4	13.9	17.6	21.5	22.8	21.6	19.2	16.8	12.9	8.3
	5%	DB	8.1	11.9	17.5	22.0	27.0	31.0	32.0	32.0	27.4	21.5	14.2	8.9
		MCWB	6.0	7.1	10.5	13.4	16.8	20.7	21.9	21.6	18.3	16.3	12.4	7.3
	10%	DB	6.9	9.9	15.6	20.0	25.2	29.2	30.8	30.2	25.8	19.8	13.0	7.8
		MCWB	5.1	6.5	9.9	12.6	16.1	20.2	21.3	20.9	17.4	15.4	11.5	6.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	10.8	14.1	17.0	21.6	26.4	26.0	25.9	23.2	20.2	16.3	10.7
		MCDB	10.3	13.9	17.6	21.3	24.9	30.2	30.4	28.7	26.0	21.7	16.6	11.4
	2%	WB	7.9	9.4	12.9	15.8	20.1	24.3	24.8	24.6	21.7	18.7	13.9	8.8
		MCDB	8.6	12.1	16.3	20.4	24.0	28.1	29.4	28.7	25.0	21.5	14.8	9.6
	5%	WB	6.5	8.2	11.9	14.8	18.9	23.0	23.9	23.6	20.6	17.6	12.9	7.7
		MCDB	7.4	10.3	15.5	19.2	23.4	28.1	28.4	27.7	24.1	20.0	13.9	8.4
	10%	WB	5.4	7.1	11.0	13.9	17.8	21.9	23.0	22.7	19.7	16.6	11.9	6.8
		MCDB	6.2	8.9	14.2	18.0	22.8	26.4	27.8	27.3	23.2	18.6	12.8	7.5
Mean Daily Temperature Range	5% DB	MDBR	7.4	9.3	10.6	10.6	10.7	10.6	11.0	10.9	10.4	8.6	6.6	6.7
		MCDBR	9.9	13.1	14.3	14.2	13.3	12.4	12.0	12.7	12.3	10.8	7.7	8.4
		MCWBR	7.0	8.0	7.0	5.8	4.8	4.7	4.5	4.6	4.5	5.4	5.6	6.2
	5% WB	MCDBR	7.7	10.5	11.4	11.9	11.7	11.4	11.0	10.9	10.4	8.5	6.1	6.0
		MCWBR	6.2	7.2	6.4	5.7	5.1	5.3	5.0	5.3	4.9	5.1	5.0	5.1
Clear-Sky Solar Irradiance	taub	0.350	0.380	0.427	0.463	0.461	0.487	0.468	0.456	0.447	0.434	0.379	0.343	
	taud	2.372	2.254	2.128	2.064	2.121	2.096	2.157	2.202	2.193	2.179	2.298	2.397	
	Ebn at Noon	741	781	791	797	813	792	802	796	764	706	689	708	
	Edh at Noon	77	103	134	156	152	156	145	134	125	110	82	68	
All-Sky Solar Radiation	RadAvg	1.42	2.23	3.54	4.50	5.57	6.38	6.53	5.57	4.12	2.48	1.47	1.21	
	RadStd	0.22	0.33	0.47	0.50	0.51	0.35	0.36	0.36	0.28	0.36	0.22	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47)	Regional (53 neighbors)	+0.34	N/A	N/A	+0.60	N/A	N/A	N/A	N/A	+65	+9		

Nomenclature: See separate page